

Title of the Project

AI-Powered Civic Issue Reporting System

Problem Statement

Urban areas frequently face civic issues such as potholes, garbage accumulation, water leakage, broken streetlights, damaged roads, and illegal dumping. Existing complaint mechanisms are often manual, fragmented, or slow, leading to delayed resolution and reduced citizen engagement. There is a need for an intelligent, user-friendly platform that enables citizens to report issues efficiently while helping authorities prioritize and manage complaints effectively.

The proposed **AI-Powered Civic Issue Reporting System** will allow users to submit complaints using images and text. Artificial Intelligence (AI) will automatically classify the reported issue, enabling faster routing to the appropriate municipal department and improving the overall complaint resolution process.

Objectives of the Project

1. Develop a web-based platform for citizens to report civic issues with images, descriptions, and locations.
2. Use AI and computer vision techniques to automatically classify reported issues (e.g., potholes, garbage, water leakage, broken streetlights).
3. Provide secure user authentication and complaint tracking.
4. Build an admin dashboard for municipal authorities to manage and update complaint statuses.
5. Generate analytics and reports to identify frequently occurring civic problems and affected areas.
6. Improve communication between citizens and local authorities through a transparent digital system.

Technology Stack

Programming Languages

- Python

- JavaScript
- HTML5
- CSS3

Frameworks

- React.js (Frontend)
- Flask or FastAPI (Backend API)
- TensorFlow/Keras (AI Model Development)

Tools

- OpenCV (Image Processing)
- Git & GitHub (Version Control)
- Postman (API Testing)
- VS Code (Development Environment)

Database

- MongoDB

Platforms & Services

- Google Maps API or OpenStreetMap (Location Services)
- Docker (Optional for Containerization)
- Vercel (Frontend Deployment)
- Render or Railway (Backend Deployment)

Domain Area of Application

- Smart Cities
- E-Governance
- Urban Infrastructure Management
- Public Service Delivery
- Artificial Intelligence and Computer Vision

Expected Outcomes

- Faster and more efficient reporting of civic issues.
- Automatic AI-based categorization of complaints, reducing manual effort.
- Improved response times through streamlined complaint management.
- Increased citizen participation in urban governance.
- Better monitoring and analysis of recurring civic issues.
- A scalable digital platform that supports smarter and more transparent municipal services.